

MvMRL: a multi-view molecular representation learning method for molecular property prediction

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Abstract

Effective molecular representation learning is very important for Artificial Intelligence-driven Drug Design because it affects the accuracy and efficiency of molecular property prediction and other molecular modeling relevant tasks. However, previous molecular representation learning studies often suffer from limitations, such as over-reliance on a single molecular representation, failure to fully capture both local and global information in molecular structure, and ineffective integration of multiscale features from different molecular representations. These limitations restrict the complete and accurate representation of molecular structure and properties, ultimately impacting the accuracy of predicting molecular properties. To this end, we propose a novel multi-view molecular representation learning method called MvMRL, which can incorporate feature information from multiple molecular representations and capture both local and global information from different views well, thus improving molecular property prediction. Specifically, MvMRL consists of four parts: a multiscale CNN-SE Simplified Molecular Input Line Entry System (SMILES) learning component and a multiscale Graph Neural Network encoder to extract local feature information and global feature information from the SMILES view and the molecular graph view, respectively; a Multi-Layer Perceptron network to capture complex non-linear relationship features from the molecular fingerprint view; and a dual cross-attention component to fuse feature information on the multi-views deeply for predicting molecular properties. We evaluate the performance of MvMRL on 11 benchmark datasets, and experimental results show that MvMRL outperforms state-of-the-art methods, indicating its rationality and effectiveness in molecular property prediction. The source code of MvMRL was released in <https://github.com/jedison-github/MvMRL>.

Keywords: molecular property prediction; multi-view learning; molecular representations; local information; global information

Introduction

Molecular property prediction has been widely considered as one of the most critical tasks in computational drug and materials discovery, as they both rely on predicted molecular properties to evaluate, select, and generate molecules [1]. However, learning molecular characterization remains the first yet vital step [2]. Molecular representation learning aims to encode input notations of molecules into numerical vectors, which later serve as features for downstream tasks such as molecular property prediction. Quantitative Structure–Activity (Property)

Relationship (QSAR/QSPR) modeling is the most widely used and well-established computational method for predicting molecular properties. Traditional QSAR/QSPR methods [3] were typically based on fixed molecular representations, such as extended-connectivity fingerprints and other chemical fingerprints. By extracting molecular structure information and converting it into numerical features, fixed molecular representations provide an important foundation for predicting molecular properties and drug design. For example, Setiya *et al.* [4] proposed MolToxPred, a machine learning tool, which uses a stacked model approach to predict the potential toxicity of small molecules. However,

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these representations heavily relied on manually crafted features, limiting their utility. Recently, the emergence and development of deep learning (DL) methods have allowed chemical information learned by large amounts of data to build a data-driven QSAR model [5]. Various molecular representations have been used as inputs in DL [6] [7], such as Simplified Molecular Input Line Entry System (SMILES) and molecular graphs.

SMILES is a widely used linear notation for representing molecular structures [1] due to its simplicity. Some DL methods for natural language processing are well suited to process SMILES. For example, Jaeger et al. [8] and Peng et al. [9] employed the concept of word2vector [10] to treat a molecule as a sequence generated by a finite number of chemical structures, and then obtain molecular features via unsupervised learning. Cao et al. [11] used a lightweight convolutional network (CNN) learning SMILES sequences for Anatomical Therapeutic Chemical classification. Xu et al. [12] applied the idea of the sequence-to-sequence learning method to convert SMILES representation learning into a text-generated reconstruction task. To further enhance SMILES-based molecular representation learning, Transformer [13] and its variants have also been utilized for molecular property prediction. For example, Honda et al. [14] proposed SMILES-Transformer, and Wang et al. [15] proposed SMILES-BERT first performed pretraining on large-scale molecular data. Besides, message passing operation on SMILES has also been proposed to capture hidden patterns in chemical strings. Wang et al. [16] proposed CLAPS, which applied contrastive learning with attention-guided positive sample selection to learn molecule representations.

A molecular graph is another representation of a molecular structure that explicitly represents the intrinsic categorical properties (e.g. atom types and bond types) and intramolecular topology of the molecule. Many graph-based methods have shown good performance on various downstream tasks. For example, Hu et al. [17] proposed a graph-based pretraining model to enhance the generalization performance of molecular graphs. Peng et al. [18] proposed MolGIN, which exploits bond features and differences in the influence of atom neighbors to end-to-end predict ADMET properties. Li et al. [19] proposed TrimNet, which utilizes a novel triplet message mechanism to learn molecular representation. Wang et al. [20] proposed a Graph attention convolutional neural network model (GACNN) with the combination of an undirected graph and attention convolutional neural network to improve the accuracy for chemical poisoning of honey bees. Recently, molecular graphs combined with Transformer have been used for molecular property prediction. Gao et al. [21] proposed TransFoxMol, which employs GNN and Transformer modules to encode local and global environments of 2D molecules. Chen et al. [22] proposed SAT, a flexible structure-aware self-attention mechanism that explicitly considers the graph structure and captures structural interaction between nodes. Jiang et al. [23] proposed PharmHGT, a pharmacophoric-constrained heterogeneous graph transformer, which obtains more informative molecular representations for molecule property prediction.

In summary, both molecular sequence-based methods and graph-based methods have shown good performance for molecular property prediction, but current molecular representation learning methods have the following limitations: (i) Different molecular representation learning describes molecular information from different perspectives. For example, molecular fingerprints and SMILES focus on chemical information, while molecular graphs emphasize spatial structural information. Therefore, combining different molecular representation methods can complement each other, providing a more comprehensive description

and understanding of the properties and behavior of molecules. (ii) The local information of a molecule includes the atom connectivity, types of chemical bonds, and properties of functional groups, which influence the chemical properties and reactivity of the molecule. Moreover, the global information of a molecule involves overall structure, stereochemistry, and spatial arrangement, which significantly impact the physical properties and biological activity of a molecule. Considering both aspects of information helps to predict and understand the properties and behavior of molecules.

Here, based on the observations above, we propose a novel method called Multi-view Molecular Representation Learning (MvMRL) to overcome the limitations of previous studies. The main contributions of this work are summarized as follows:

- MvMRL integrates multiple molecular representations from multi-views, including molecular fingerprints, molecular graphs, and SMILES, which allows for more complete and accurate representations of molecular structures and properties.
- This work develops two multiscale feature extractors specifically designed for molecular SMILES and molecular graphs, respectively, to better learn the local and global information of molecules.
- This work develops a dual cross-attention component in the multi-view feature fusion module, which enables the model to concentrate on the crucial features between multiscale and nonlinear relationships.

The methodology of MvMRL

In this section, we will first summarize the proposed multi-view learning framework MvMRL (Section Overview of MvMRL). Then, we will introduce more details of three main modules of MvMRL including the multi-view feature learning module (Sections Multiscale CNN-SE for SMILES feature learning, MLP for molecular fingerprint feature learning, and Multiscale GNN encoder for molecular graph feature learning), multi-view feature fusion module (Section Multi-view feature fusion), and prediction (Section Prediction), respectively.

Overview of MvMRL

Figure 1A depicts the overall architecture of our MvMRL, which consists of three main modules: multi-view feature learning module, multi-view feature fusion module, and prediction module. Specifically, the multi-view feature learning module consists of a multiscale CNN-SE SMILES learning component, a multiscale GNN encoder, and a multilayer perceptron (MLP). This module is designed to capture local and global feature information from the SMILES view and the molecular graph view while generating more abstract and informative feature information on the molecular fingerprints. The multi-view fusion module utilizes a dual cross-attention component to interact with feature information from three views of molecules. The prediction module aims to capture high-level feature relationships and predict molecular properties.

Multiscale CNN-SE for SMILES feature learning

CNN networks learn text features by fusing spatial correlations between features, benefiting from the convolutional kernel's local receptive field but also facing limitations imposed by it. The squeeze-and-excitation (SE) block [24] can adaptively adjust the weights of channel features, allowing the network to focus more

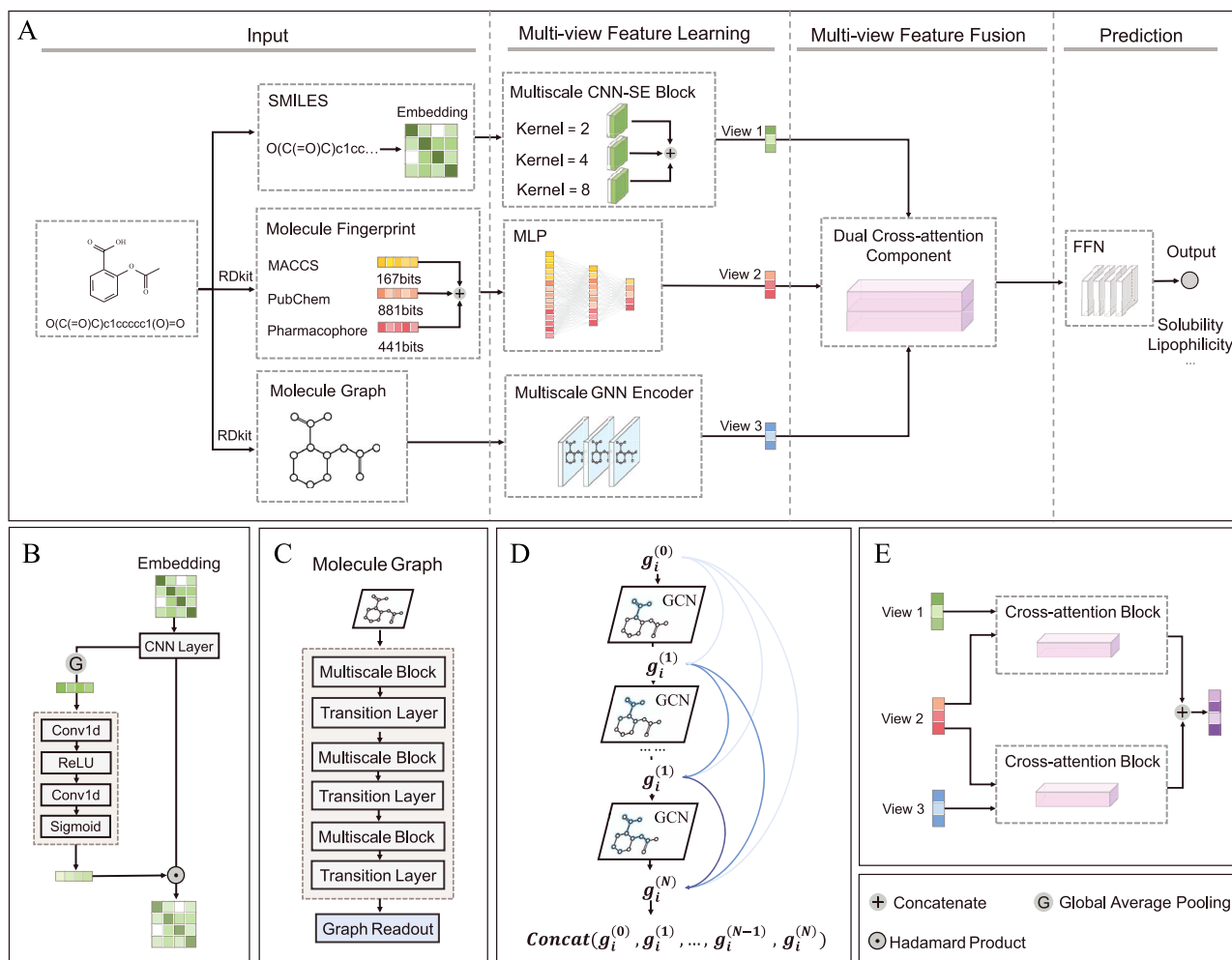


Figure 1. The overview of the MvMRL framework and its detailed structure; (A) illustrates the overall workflow of MvMRL architecture; (B) shows the detail of the multiscale CNN-SE SMILES learning component; (C) is the workflow of the multiscale GNN encoder; (D) Describes the detail of the multiscale block in the multiscale GNN encoder; (E) shows the detail of the dual cross-attention component.

on important features [25]. In this work, we developed a multi-scale CNN-SE SMILES learning component to better learn SMILES features.

The embedding for SMILES sequences

To facilitate CNNs to learn the features of the SMILES sequences, we first built dictionaries to encode each character in the sequence as a token. Next, a SMILES sequence was mapped into a token sequence. For example, the SMILES sequence of Propylene glycol is “CC(O)CO,” which can be encoded as a vector $Vec = [43, 43, 1, 49, 32, 43, 49]$ based on the SMILES dictionary {“(:1, ‘)’”: 32, “C”:43, “O”:49}, as shown in Fig. 2. Then, every token in the sequence was encoded to generate an embedding vector S_e by the SMILES embedding layer. Finally, all the embedding vectors are merged into an embedding matrix that represents the SMILES sequence as the input for the multiscale CNN-SE SMILES learning component.

The squeeze-and-excitation block

SE block consists of squeeze, excitation, and reweighting operators. Given a feature matrix U as the output of the convolution layer, it was sent into the corresponding SE block. The squeeze operator aims to transform the dimensions of the feature matrix U and obtains the channel-wise statistic $z \in \mathbb{R}^C$ by applying the

global average pooling operation, as follows:

$$z = F_{sq}(U) = \text{AvgPool}(U) \quad (1)$$

where $\text{AvgPool}(\cdot)$ is the global average pooling. The excitation operator uses two learnable fully connected layers to obtain the weights of each channel, and uses the weighted features as an input to the next layer of the network s , which is computed as follows:

$$s = F_{ex}(z, W) = \sigma(W_2 \text{Conv1d}(\delta(W_1 \text{Conv1d}(z)))) \quad (2)$$

where $F_{ex}(\cdot)$ is the excitation operator, $\sigma(\cdot)$ is the sigmoid function, $\delta(\cdot)$ is the Rectified Linear Unit (ReLU) activation function, $\text{Conv1d}(\cdot)$ is the 1D convolutional operator. $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ are the two learnable weight matrices, respectively, in which the reduction ratio r is set to balance between performance and complexity. After obtaining the weighted feature s , the reweighted representation x is obtained by performing the Hadamard product, as follows:

$$x = SE(U, s) = U \odot F_{sq}(U, W) \quad (3)$$

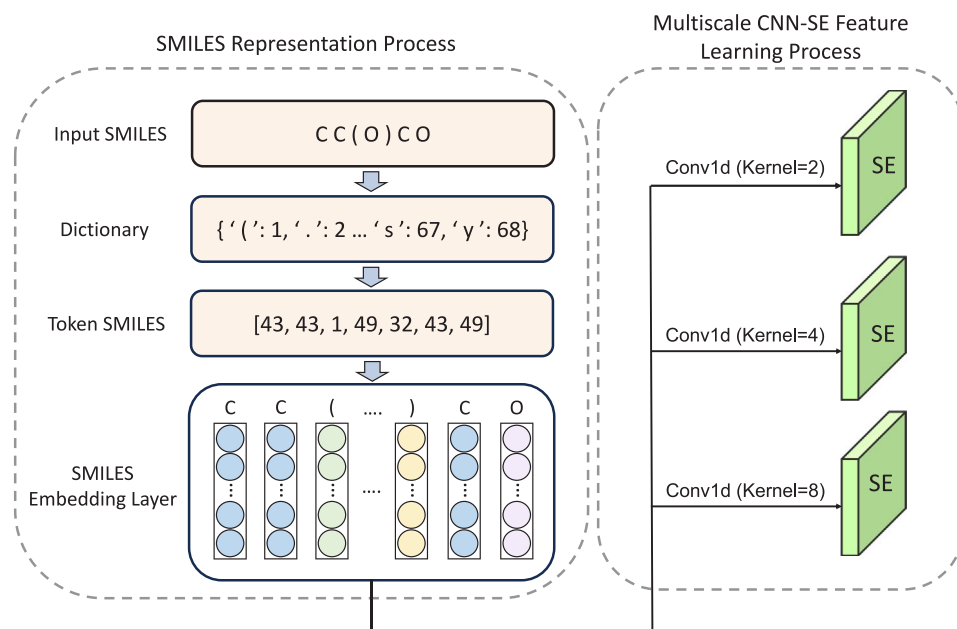


Figure 2. The architecture of the multiscale CNN-SE SMILES learning component.

Multiscale CNN-SE SMILES learning component

SMILES sequences contain semantic information at different scales, and a single-scale CNN-SE module may not fully capture the multiscale feature information in SMILES sequences. Therefore, in this work, we proposed a multiscale CNN-SE SMILES learning component, which can not only capture semantic information of the SMILES sequences at different scales but also consider the spatial correlation between features.

The whole process of the multiscale CNN-SE SMILES learning component mainly includes two parts: the SMILES representation process and the multiscale CNN-SE SMILES feature learning process, as shown in subfigures (A) and (B) of Fig. 1. First, the multiscale CNN-SE SMILES learning component inputs the SMILES embedding S_e into the convolution layer with different kernel sizes for extracting features at different scales, and then feeds the extracted features into the corresponding SE block. In the SE block, a reweighted weight matrix is obtained by Equation (3). Finally, features from different scale CNN-SEs are concatenated. Hence, the molecular SMILES representation h_{SM} can be formulated as follows:

$$h_{SM} = \text{Concat}(\text{SE}(\text{CNN}_{K=k}(S_e))) \quad (4)$$

where $\text{CNN}_{K=k}(\cdot)$ is the 1D convolutional operator. K is the size of the convolutional kernel, and the value of k is set as 2, 4, and 8, respectively. $\text{Concat}(\cdot)$ is the concatenate operation.

MLP for molecular fingerprint feature learning

The molecular fingerprint is a molecular representation that provides specific information about the molecular structure. A single molecular fingerprint is difficult to capture all the important structures or properties of a molecule. In this work, we followed the literature [22] in using a combination of three different fingerprints: the MACCS fingerprint, the Pharmacophore ErG fingerprint, and the PubChem fingerprint to better learn features from the molecular fingerprint view. Each fingerprint is a fixed-length binary vector representation for each molecule, with each bit in the vector corresponding to a unique molecular substructure. The value of the corresponding bit in the fingerprint vector is

determined by the presence or absence of a substructure in a molecule. The MACCS fingerprint can capture common chemical features for molecules such as aromatic rings and ring structures [26]. The PubChem fingerprint can capture information about the topology, bonds, and substructures for molecular description and similarity analysis [27]. The Pharmacophore ErG fingerprint is a 2D pharmacophore fingerprint using chemical information and structural information, where chemical information refers to a set of rules for specifying atom types defined using the SMARTS pattern [28].

Specifically, we first apply fingerprint encoding methods to obtain three binary vector representations, including PubChem fingerprint FP_{Pub} , MACCS fingerprint FP_{MAC} , and Pharmacophore ErG fingerprint FP_{Phar} . Then, these three fingerprint vectors are concatenated to obtain a mixed fingerprint FP , as shown in Equation (5). Finally, an MLP network, which can approximate any continuous mapping, is used to capture complex nonlinear relationship features h_{FP} from the mixed fingerprint FP , as shown in Equation (6).

$$FP = \text{Concat}(FP_{Pub}, FP_{MAC}, FP_{Phar}) \quad (5)$$

$$h_{FP} = \text{MLP}(FP) \quad (6)$$

Multiscale GNN encoder for molecular graph feature learning

Molecular graph representations

A molecule can be represented as an undirected graph, denoted as $G = (V, E)$, where V represents a set of the nodes (atoms) and $E \subseteq V \times V$ represents a set of the edges (bonds). As shown in Fig. 3A, a molecular graph can be constructed by the open-source molecular processing toolkit RDKit [29]. Following HiGNN [30], atomic attributes such as atomic number, hybridization, formal charge, and aromaticity are used for each node (atom) in this work. In addition, bond attributes such as bond type and conjugation are used for the properties of each edge (bond).

GNNs and their variants have been used to learn molecular graph representations. GNNs map a molecular graph G to its

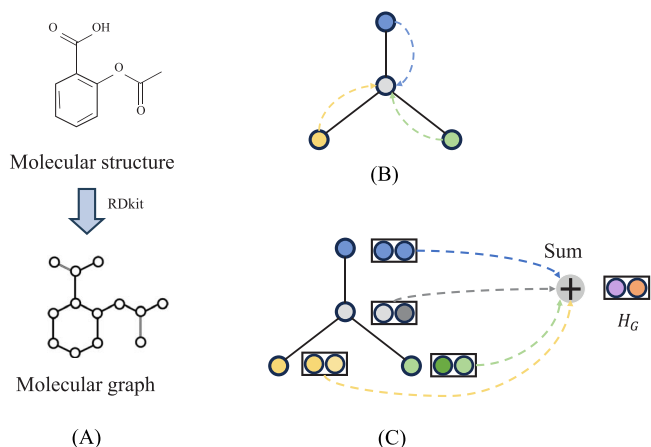


Figure 3. Molecular graph representation and graph embedding; (A) illustrates that molecular structure converts to molecular graph by RDKit; (B) and (C) are the message passing phase and graph readout phase in GNN, respectively.

corresponding graph representation $h_G \in \mathbb{R}^d$ via a message passing phase and a readout phase, as shown in subfigures (b) and (c) of Fig. 3. However, shallow GNNs have limitations in capturing the global structure of molecules, while a well-constructed GNN can preserve the local structure and global structure of molecules. In this work, we used Multiscale GNN [31] to learn both local and global features from the molecular graph view. The Multiscale GNN is composed of two parts: the multiscale block and the transition layer.

Multiscale block

The multiscale block is employed to increase the depth of the network and extract multiscale feature information. As shown in Fig. 1C, a multiscale block contains n graph convolutional layers. Specifically, for given a molecular graph G , the i -th vertex embedding at the step t is denoted as $g_i^{(t)}$. Then, the vertex embedding $g_i^{(t+1)}$ at the time step $t+1$ is represented as:

$$g_i^{(t+1)} = \delta \left(\text{BN} \left(W_4 x_i^{(t)} + W_5 \sum_{j \in N(i)} x_j^{(t)} \right) \right) \quad (7)$$

where W_4 and W_5 are learned parameters, $N(i)$ is the set of neighbors of a vertex i . $\text{BN}(\cdot)$ is a node-level batch normalization followed by a ReLU activation function, which is indispensable for achieving optimal performance in very deep models. Furthermore, multiscale blocks utilize dense connections to increase the depth of the network, effectively alleviating the vanishing gradient problem. The multiscale block can be represented as:

$$g_i^{(1)} = H(g_i^{(0)}, \theta_1) \quad (8)$$

$$g_i^{(2)} = H(\text{Concat}(g_i^{(0)}, g_i^{(1)}), \theta_2) \quad (9)$$

$$g_i^{(3)} = H(\text{Concat}(g_i^{(0)}, g_i^{(1)}, g_i^{(2)}), \theta_3) \quad (10)$$

$$g_i^{(n)} = H(\text{Concat}(g_i^{(0)}, g_i^{(1)}, \dots, g_i^{(n-1)}), \theta_n) \quad (11)$$

where $H(\cdot)$ is the graph convolutional operation described by Equation (7) with parameters θ_n (consists of W_1 and W_2). n represents n -th graph convolution operation.

Transition layers

The transition layers are used to integrate the multiscale features extracted from the previous multiscale blocks and reduce the number of channels in the feature maps, as shown in Fig. 1C.

Assuming that the input multiscale feature with time step $t+1$ is $\text{Concat}(g_i^{(0)}, g_i^{(1)}, \dots, g_i^{(t)})$, the transition layer can be represented as:

$$g_i^{(t+1)} = \text{BN} \left(\Phi_1 \left(\text{Concat}(g_i^{(0)}, g_i^{(1)}, \dots, g_i^{(t)}) \right) + \Phi_2 \sum_{j \in N(i)} \left(\text{Concat}(g_j^{(0)}, g_j^{(1)}, \dots, g_j^{(t)}) \right) \right) \quad (12)$$

where Φ_1 and Φ_2 are learnable weight matrices shared across all vertices. Finally, the readout layer described by Equation (13) is used to convert the whole graph into a feature vector h_G to aggregate the vertex embedding into a graph embedding.

$$h_G = \frac{1}{|V|} \sum_{v \in V} g_v^{(L)} \quad (13)$$

where $|V|$ is the number of vertices in the molecular graph, L is the final time step.

Multi-view feature fusion

The main purpose of feature fusion is to extract and combine diverse complementary feature information from different views. The cross-attention mechanism helps the model understand the correlations and importance of different features, leading to better integration of multi-view feature information. In this module, we proposed a dual cross-attention component for feature fusion, capturing multiscale feature information and enhancing feature representation capability. The dual cross-attention component mainly consists of a cross-attention block between the multiscale CNN-SE SMILES learning component and the MLP network, a cross-attention block between the Multiscale GNN encoder and the MLP network, as well as a concatenation layer, as shown in Fig. 1.

Specifically, after obtaining the output h_G , h_{FP} , and h_{SM} from the previous module, we combine them using the dual cross-attention component. First, we construct learnable linear transition layers that make a linear projection of h_{SM} , h_{FP} , and h_G as query, key, and value, respectively.

$$Q_{SM} = f(h_{SM} W_{SMq} + b_{SMq}), Q_{FP} = f(h_{FP} W_{FPq} + b_{FPq}), \\ Q_G = f(h_G W_{Gq} + b_{Gq}) \quad (14)$$

$$K_{SM} = f(h_{SM} W_{SMk} + b_{SMk}), K_{FP} = f(h_{FP} W_{FPk} + b_{FPk}), \\ K_G = f(h_G W_{Gk} + b_{Gk}) \quad (15)$$

$$V_{SM} = f(h_{SM} W_{SMv} + b_{SMv}), V_{FP} = f(h_{FP} W_{FPv} + b_{FPv}), \\ V_G = f(h_G W_{Gv} + b_{Gv}) \quad (16)$$

where W_{SMq} and b_{SMq} are weight and bias, respectively, and $\{Q_{SM}, K_{SM}, V_{SM}\}$ is generated by the projection function $f(\cdot)$. Similarly, $\{Q_{FP}, K_{FP}, V_{FP}\}$ and $\{Q_G, K_G, V_G\}$ can be obtained. Then, one of the cross-attention coefficients Attn_{SF} can be obtained as follows:

$$\text{Attn}_{SF1} = \text{CrossAttention}(Q_{SM}, K_{SM}, V_{FP}) \\ = \text{Softmax} \left(\frac{Q_{SM} K_{SM}^T}{\sqrt{C/d}} \right) \cdot V_{FP} \quad (17)$$

$$\text{Attn}_{SF2} = \text{CrossAttention}(Q_{FP}, K_{FP}, V_{SM}) \\ = \text{Softmax} \left(\frac{Q_{FP} K_{FP}^T}{\sqrt{C/d}} \right) \cdot V_{SM} \quad (18)$$

$$\text{Attn}_{SF} = \text{Concat}(\text{Attn}_{SF1}, \text{Attn}_{SF2}) \quad (19)$$

where $Attn_{SF1}$ and $Attn_{SF2}$ are the corresponding attention coefficients with $\{Q_{SM}, K_{SM}, V_{SM}\}$ and $\{Q_{FP}, K_{FP}, V_{FP}\}$ as inputs. C and d are the embedding dimensions and number of heads, respectively. Next, another cross-attention coefficient $Attn_{GF}$ can be formulated as follows:

$$Attn_{GF1} = \text{CrossAttention}(Q_G, K_G, V_{FP}) = \text{Softmax}\left(\frac{Q_G K_G^T}{\sqrt{C/d}}\right) \cdot V_{FP} \quad (20)$$

$$Attn_{GF2} = \text{CrossAttention}(Q_{FP}, K_{FP}, V_G) = \text{Softmax}\left(\frac{Q_{FP} K_{FP}^T}{\sqrt{C/d}}\right) \cdot V_G \quad (21)$$

$$Attn_{GF} = \text{Concat}(Attn_{GF1}, Attn_{GF2}) \quad (22)$$

Finally, the attention coefficients $Attn_{SF}$ and $Attn_{GF}$ are concatenated to obtain the fusion feature $Attn$.

$$Attn = \text{Concat}(Attn_{SF}, Attn_{GF}) \quad (23)$$

Prediction

As shown in Fig. 1A, the output $Attn$ of the dual cross-attention component is fed into the Feed Forward Network (FFN) to predict molecular property, which can be formulated as Equation (24). Concretely, the FFN contains three linear transformation layers, each linear transformation layer is followed by a ReLU activation and dropout layer with a dropout rate of 0.1.

$$\text{Output} = \text{FFN}(Attn) \quad (24)$$

where Output is the predicted result of the classification/regression task.

Experiments

In this section, we evaluate MvMRL against state-of-the-art methods on 11 public benchmark datasets. We will first provide a brief introduction for these datasets and baselines. Then, we will introduce the experimental setting. Finally, we will present the experimental results and analyze the ablation experiments.

Benchmark datasets

To evaluate the prediction capability of our MvMRL method, we conduct experiments on 11 benchmark datasets including eight classification datasets and three regression datasets. Among the benchmark datasets, there are eight datasets from MoleculeNet [32], including the BBBP, BACE, ClinTox, Tox21, and Sider datasets for the classification task, and the FreeSolv, ESOL, and Lipo datasets for the regression task. In addition, there are three small-scale datasets including DHFR, IGF1R, and AHR from ChEMBL [21] for assessing the model's classification performance on small sample data. Table 1 summarizes the statistics of the datasets used in the experiments. Details of each dataset are listed as follows.

- BBBP contains compounds labeled as either blood-brain barrier permeable or non-permeable.
- BACE contains qualitative binding results of a group of human β -secretase one inhibitor. It provides the correlations between molecules and their corresponding biological activity information.

- ClinTox contains drugs approved by the FDA, and drugs that failed clinical trials for toxicity reasons.
- Tox21 contains large amounts of experimental compounds for 12 different target relevant to drug toxicity.
- Sider contains marked drugs along with side effects and these drugs are grouped into 27 system organ classes.
- FreeSolv contains compounds with their logarithmic solubility values.
- ESOL contains compounds with their water solubility values.
- Lipo contains compounds with their octanol/water distribution coefficient values.
- DHFR, IGF1R, and AHR are three small bio-relevant datasets with molecule counts between 124 and 165, uniformly distributed and with balanced data labels.

Baselines

We conducted a comprehensive evaluation of MvMRL against 24 state-of-the-art methods to fully demonstrate the superiority of our MvMRL. The baselines contain SMILES sequence-based models, molecular graph-based models, and the models using the feature combination of SMILES sequence and molecular graph. Specifically, Transformer [13], ST (SMILES Transformer) [14], RNN [33], and RNNS2S+MLP [12] are SMILES sequence-based DL model using the SMILES strings as the model input. TrimNet [19], TransFoxMol [21], FP-GNN [34], HiMol [6], GCN [35], Weave [36], MPNN [37], N-Gram [38], DMPNN [39], GraphConv [40], AttentiveFP [41], PretrainGNN [42], and HimGNN [43] are graph neural network methods, where HiMol [6] introduces graph self-supervised learning. GROVER [44] follows the message-passing methods operated on directed graphs, while MoCL [45], MolCLR [46], and GraphCL [47] introduce contrastive learning. GEM [1] and GraphMVP [48] are self-supervised learning models that incorporate molecular 3D geometry information. GraSeq [49] mainly focuses on combining molecular SMILES and graph representations to predict molecular properties.

Experimental settings

Dataset partitioning

We split all the datasets using two ways: random splitting and scaffold splitting, respectively. The random splitting randomly splits molecules in datasets into train, validation, and test sets. Scaffold splitting splits molecules based on their substructure, which is a more challenging way of splitting the data. Both two splitting ways were adopted with a ratio for train/validation/test sets as 0.8:0.1:0.1.

Evaluation method

We performed 10 independent runs with different random seeds on two types of splitting ways and calculated the mean and standard deviation values of ROC-AUC or RMSE.

Hyperparameter setting

Table 2 demonstrates the main hyperparameter settings for MvMRL used in our experiments. We optimized the learning rate and weight decay hyperparameters using the Tree Parzen Estimator algorithm implemented in the Hyperopt Python packages [50]. The final learning rate and weight decay vary from dataset to dataset.

Runtime environment

Our experiments are run on a GPU server with an NVIDIA GeForce RTX 3090 graphics card with 24GB of memory.

Table 1. Statistics of the datasets used in the experiments

Category	Dataset	#Tasks	#samples	Type	Metric
Physiology	BBBP	1	2040	Classification	ROC-AUC
	ClinTox	2	1478	Classification	ROC-AUC
	Tox21	12	7831	Classification	ROC-AUC
	Sider	27	1427	Classification	ROC-AUC
Biophysics	BACE	1	1513	Classification	ROC-AUC
Physical chemistry	FreeSolv	1	639	Regression	RMSE
	ESOL	1	1127	Regression	RMSE
	Lipo	1	4200	Regression	RMSE
	DHFR	1	165	Classification	ROC-AUC
Other (small dataset)	IGF1R	1	157	Classification	ROC-AUC
	AHR	1	124	Classification	ROC-AUC

Table 2. Main hyperparameters used in MvMRL

Name of hyperparameter	Value
Size of the convolution kernel	2, 4, 8
Multiscale block	3
Batch size	64
Early stop	50
Max epochs	200
Optimizer	Adam
Learning rate	$[\log(10^{-4}), \log(10^{-2})]^a$
Weight decay	$\{10^{-4}, 10^{-5}, 10^{-6}\}$

^aReturns a value drawn according to $\exp(\text{uniform}(\text{low}, \text{high}))$ so that the logarithm of the return value is uniformly distributed.

Results and analysis

Comparison with the baselines

Tables 3–5 summarize the experimental results of the comparative models for the classification (or regression) tasks on random splitting and scaffold splitting, respectively. The best and the second-best performances are highlighted in bold and italicized, respectively.

Results on random splitting

Table 3 shows that our MvMRL outperforms all state-of-the-art results on the 5/8 datasets and ranks second on the remaining three datasets for classification tasks. The area under the receiver operating characteristic (ROC) curve (AUC) values obtained by MvMRL have improved by 0.7%, 1.0%, 2.1%, and 0.1% compared with the state-of-the-art results on the BBBP, BACE, ClinTox, and Tox21 datasets, respectively. For small-scale datasets, MvMRL outperforms all state-of-the-art results on the 1/3 dataset and ranks second on other two datasets. It demonstrates that MvMRL has some effectiveness in predicting small-scale datasets. For regression tasks, Table 4 shows that our MvMRL outperforms all state-of-the-art results on the 2/3 dataset and wins the second-best on the remaining dataset, Lipo, on which our MvMRL only underperforms FP-GNN. This may be because FP-GNN specifically focuses on the substructural information of molecules, having more effects on the Lipo dataset. The Root Mean Square Error (RMSE) values obtained by MvMRL have improved by 3.3% and 5.1% compared with the state-of-the-art results on the FreeSolv and ESOL datasets, respectively.

Results on scaffold splitting

When using the scaffold splitting method, our MvMRL outperforms the state-of-the-art results on the 5/8 benchmark datasets,

as shown in Table 5. Note that in accordance with TransFoxMol [21], we did not conduct comparative experiment on the small-scale datasets including DHFR, IGF1R, and AHR because no studies have reported experiments about the small-scale datasets on scaffold splitting. For classification tasks, the ROC-AUC values obtained by MvMRL have improved by 3.5% and 5.9% compared with the state-of-the-art results on the BBBP and ClinTox datasets, respectively. For regression tasks, the RMSE values obtained by MvMRL have improved by 0.3% and 1.3% compared with the state-of-the-art results on the FreeSolv and ESOL datasets, respectively.

In summary, compared with the state-of-the-art methods based on SMILES or graphs or the combination of SMILES and graphs, our MvMRL has achieved better performance on most of the benchmark datasets. This indicates that MvMRL can better capture the commonalities and differences between the graph and textual views, and leverage the advantages of multi-view information of the molecule.

Ablation study

To investigate the contribution of the main components in MvMRL, we conducted ablation experiments using random splitting on the six benchmark datasets. The ablation studies include exploring the effects of different components, different types of molecular fingerprints, different sizes of convolution kernels, and different module aggregation methods.

Analysis of ablation studies on the architecture components of MvMRL

To better understand the effect of different components in our MvMRL, we first designed 12 variants, each consisting of one or more essential modules (components). Then, we conducted comparative experiments on these variants. Table 6 summarizes these experimental results.

As shown in Table 6, it can be found that MvMRL achieves better results in five out of six datasets compared to other variants, which clearly illustrates the effectiveness of the inference process in establishing a correlation between molecular feature information and target properties. The comparative results provide the following observations: (a) single-view multiscale feature extractors can capture features at different levels, but multi-view feature extractors are more comprehensive in capturing the diversity and complexity of molecules. Simultaneously, multiscale feature extraction in multi-views captures local and global features better than single-scale feature extraction. (b) Combining the features extracted from three views (SMILES sequences, molecular graph and molecular fingerprints) and

Table 3. The performance comparison on classification tasks, utilizing the random splitting

Model	BBBP	BACE	Clintox	Tox21	Sider	DHFR	IGF1R	AHR
RNN [51]	0.902 ± 0.015	–	0.915 ± 0.009	0.806 ± 0.007	0.605 ± 0.009	–	–	–
Transformer [51]	0.944 ± 0.011	–	0.954 ± 0.003	0.813 ± 0.013	0.602 ± 0.012	–	–	–
GCN [51]	0.69	–	0.807	0.829	0.638	–	–	–
Weave [51]	0.671	–	0.832	0.82	0.581	–	–	–
MPNN [51]	0.910 ± 0.032	–	0.881 ± 0.037	0.844 ± 0.014	0.641 ± 0.014	–	–	–
N-Gram [51]	0.912 ± 0.013	–	0.855 ± 0.037	0.842	–	–	–	–
GROVER [51]	0.955 ± 0.003	–	0.929 ± 0.178	0.842 ± 0.009	0.663 ± 0.008	–	–	–
TrimNet [19]	0.889 ± 0.020	0.841 ± 0.043	0.948	–	–	–	–	–
FP-GNN [28]	0.935	0.881	0.84	0.815	0.661	–	–	–
MoCL [45, 51]	0.911	–	0.75	0.824	0.628	–	–	–
TransFoxMol [21]	0.919 ± 0.011	0.882 ± 0.014	0.942 ± 0.043	0.823 ± 0.008	0.651 ± 0.013	0.927 ± 0.027	0.809 ± 0.116	0.690 ± 0.098
Groverbase [21]	0.885 ± 0.027	0.853 ± 0.010	0.860 ± 0.014	0.797 ± 0.012	0.606 ± 0.056	0.871 ± 0.055	0.667 ± 0.139	0.676 ± 0.305
AttentiveFP [21]	0.901 ± 0.021	0.846 ± 0.012	0.938 ± 0.048	0.815 ± 0.012	0.606 ± 0.025	0.852 ± 0.027	0.637 ± 0.100	0.654 ± 0.157
GEM [21]	0.902 ± 0.023	0.877 ± 0.037	0.886 ± 0.014	0.844 ± 0.005	0.640 ± 0.031	0.867 ± 0.040	0.664 ± 0.100	0.606 ± 0.014
PretrainGNN [21]	0.893 ± 0.018	0.843 ± 0.019	0.693 ± 0.060	0.801 ± 0.016	0.620 ± 0.017	0.812 ± 0.070	0.647 ± 0.135	0.555 ± 0.216
DMPNN [21]	0.918 ± 0.002	0.798 ± 0.036	0.857 ± 0.029	0.821 ± 0.010	0.568 ± 0.020	0.830 ± 0.069	0.656 ± 0.043	0.522 ± 0.027
GraphMVP [21]	0.900 ± 0.036	0.869 ± 0.012	0.790 ± 0.060	0.803 ± 0.016	0.647 ± 0.021	0.837 ± 0.080	0.597 ± 0.188	0.454 ± 0.160
MolCLR [21]	0.894 ± 0.036	0.822 ± 0.044	0.856 ± 0.036	0.822 ± 0.028	0.627 ± 0.042	0.867 ± 0.072	0.634 ± 0.158	0.538 ± 0.118
GraSeq [51]	0.942 ± 0.012	–	0.918 ± 0.005	0.81	0.62	–	–	–
MvMRL(ours)	0.962 ± 0.011	0.891 ± 0.019	0.975 ± 0.019	0.845 ± 0.011	0.653 ± 0.027	0.961 ± 0.043	0.670 ± 0.175	0.673 ± 0.178

Note: The best and second-best performances are bolded and italicized, respectively. “–” means the results were not reported in the references.

Table 4. The performance comparison on regression tasks, utilizing the random splitting

Model	Freesolv	ESOL	Lipo
RNN [51]	1.108 ± 0.146	0.743 ± 0.020	0.770 ± 0.025
Transformer [51]	1.021 ± 0.102	0.767 ± 0.079	0.900 ± 0.023
GCN [51]	1.4	0.97	–
Weave [51]	1.22	0.61	–
MPNN [51]	1.242 ± 0.249	0.702 ± 0.042	0.645 ± 0.075
N-Gram [51]	2.512 ± 0.190	1.100 ± 0.160	0.876 ± 0.033
GROVER [51]	1.987 ± 0.072	0.911 ± 0.116	0.643 ± 0.030
TrimNet [19]	1.639 ± 0.406	0.770 ± 0.071	1.202 ± 0.032
FP-GNN [28]	0.905	0.675	0.625
MoCL [45, 51]	1.478	0.934	0.742
GraSeq [51]	0.865 ± 0.032	0.652 ± 0.039	0.648 ± 0.041
MvMRL (ours)	0.832 ± 0.128	0.601 ± 0.055	0.634 ± 0.018

using a good attention mechanism (the dual cross-attention component) enables a more comprehensive understanding of the relationships between features of different views, and adapts to the varying importance of the features of different views.

In summary, the superiority of MvMRL can be attributed to its ability to capture significant local and global information, topological structure information, and atomic sequence information from molecular graphs, fingerprints, and SMILES sequences. This work leverages multi-view information to provide a more precise and comprehensive representation of molecules.

Exploring the effects of different types of molecular fingerprints

In this section, we investigate MvMRL using three molecular fingerprints and a combination of these three fingerprints, respectively, to explore the impact of different types of molecular fingerprints on MvMRL’s performance. As illustrated in Fig. 4, on all datasets, the three different fingerprints underperform

the combination of the three fingerprints, respectively, indicating that these different fingerprints are complementary in capturing different molecular features. Specifically, the MACCS fingerprint captures the structural features of molecules, the PubChem fingerprint contains the topological structure and atomic properties of molecules, and the Pharmacophore ErG fingerprint captures the pharmacological features and structural information of molecules. This combination coding of the three fingerprints enables a comprehensive description of the structural and biological activity information of molecules. Overall, the experimental results demonstrate that the combination of MACCS, PubChem, and Pharmacophore ErG fingerprints is effective and advantageous for predicting molecular properties.

Exploring the effects of different sizes of convolutional kernel

To investigate the impact of different sizes of convolutional kernel of MvMRL on molecular property prediction, we evaluated the

Table 5. The performance comparison, utilizing the scaffold splitting

Model	BBBP	BACE	ClinTox	Tox21	Sider	FreeSolv	ESOL	Lipo
RNN [51]	0.832 ± 0.015	–	0.879 ± 0.009	0.725 ± 0.017	0.593 ± 0.010	2.232 ± 0.299	0.952 ± 0.034	0.891 ± 0.009
Transformer [51]	0.900 ± 0.053	–	0.905 ± 0.064	0.706 ± 0.021	0.559 ± 0.017	2.246 ± 0.237	1.144 ± 0.118	1.169 ± 0.031
ST [14]	0.9	0.719	0.963	0.706	0.559	2.246	1.144	1.169
RNNS2S + MLP [16]	0.884	0.717	0.904	0.702	0.558	2.987	1.317	1.219
GCN [51]	0.877 ± 0.036	–	0.845 ± 0.051	0.772 ± 0.041	0.593 ± 0.035	2.900 ± 0.135	1.068 ± 0.050	0.712 ± 0.049
Weave [51]	0.837 ± 0.065	–	0.823 ± 0.023	0.741 ± 0.044	0.543 ± 0.034	2.398 ± 0.250	1.158 ± 0.055	0.813 ± 0.042
MPNN [51]	0.913 ± 0.041	–	0.879 ± 0.054	0.808 ± 0.024	0.595 ± 0.030	2.185 ± 0.952	1.167 ± 0.430	0.672 ± 0.051
N-Gram [51]	0.912 ± 0.013	–	0.855 ± 0.037	0.769 ± 0.027	0.632 ± 0.005	2.512 ± 0.190	1.100 ± 0.160	0.876 ± 0.033
GROVER [51]	0.911 ± 0.008	–	0.884 ± 0.013	0.803 ± 0.020	0.624 ± 0.006	1.987 ± 0.072	0.911 ± 0.116	0.643 ± 0.030
TrimNet [19]	0.85	0.878	0.906	0.812	0.606	2.529	1.282	0.702
FP-GNN [5]	0.910 ± 0.027	0.845 ± 0.028	0.765 ± 0.038	0.803 ± 0.024	0.598 ± 0.014	2.492 ± 0.649	1.282 ± 0.332	0.679 ± 0.152
HiMol [36]	0.732	0.843	0.808	0.760 ± 0.020	0.613 ± 0.050	2.283	0.938	0.708
MolCLR [46]	0.738 ± 0.020	0.788 ± 0.050	0.867 ± 0.010	0.747 ± 0.080	0.669 ± 0.012	2.39 ± 0.14	1.16 ± 0.00	0.78 ± 0.01
GraphCL [47]	0.696	0.753	0.759	0.738	0.605	–	–	–
GraphConv [16]	0.795	0.744	0.936	0.687	0.557	3.476	1.673	1.062
HimGNN [5]	0.928 ± 0.027	0.856 ± 0.034	0.917 ± 0.030	0.807 ± 0.017	0.642 ± 0.023	1.921 ± 0.474	0.870 ± 0.154	0.632 ± 0.016
TransFoxMol [21]	0.925 ± 0.018	0.865 ± 0.005	0.972 ± 0.024	0.804 ± 0.010	0.629 ± 0.024	1.773 ± 0.368	0.869 ± 0.035	0.633 ± 0.016
AttentiveFP [21]	0.832 ± 0.010	0.782 ± 0.053	0.941 ± 0.053	0.802 ± 0.008	0.576 ± 0.050	1.838 ± 0.436	0.887 ± 0.040	0.654 ± 0.016
GEM [21]	0.897 ± 0.037	0.859 ± 0.052	0.862 ± 0.041	0.820 ± 0.008	0.626 ± 0.034	2.046 ± 0.647	0.870 ± 0.040	0.672 ± 0.020
PretrainGNN [21]	0.865 ± 0.039	0.818 ± 0.018	0.560 ± 0.049	0.770 ± 0.011	0.615 ± 0.039	3.250 ± 0.979	1.371 ± 0.026	0.779 ± 0.069
DMPNN [21]	0.867 ± 0.033	0.798 ± 0.036	0.857 ± 0.029	0.784 ± 0.009	0.568 ± 0.020	2.193 ± 0.706	0.949 ± 0.128	0.673 ± 0.043
GraphMVP [21]	0.863 ± 0.053	0.813 ± 0.022	0.560 ± 0.058	0.775 ± 0.013	0.604 ± 0.048	2.860 ± 0.948	1.431 ± 0.220	0.694 ± 0.043
GraSeq [51]	0.926 ± 0.011	–	0.904 ± 0.004	0.755 ± 0.004	0.609 ± 0.009	2.218 ± 0.184	1.034 ± 0.035	0.780 ± 0.012
MvMRL(ours)	0.963 ± 0.019	0.878 ± 0.336	0.976 ± 0.013	0.794 ± 0.014	0.607 ± 0.023	1.770 ± 0.229	0.856 ± 0.088	0.686 ± 0.037

Note: The best and second-best performances are bolded and italicized, respectively. “–” means the results were not reported in the references.

Table 6. Ablation study results of MvMRL and the variants on random splitting

Module selection						Classification (ROC-AUC)			Regression (RMSE)			
View1		View2	View3		Fusion method		BBBP	BACE	ClinTox	Free-Solv	ESOL	Lipo
Single	Multi ^c		GCN	Multi ^g	Concat	ATT						
	✓						0.945	0.818	0.975	2.287	0.847	0.997
		✓					0.926	0.881	0.793	0.873	0.682	0.676
				✓			0.914	0.884	0.883	1.239	0.786	0.637
	✓			✓	✓		0.954	0.889	0.968	1.161	0.665	0.686
	✓			✓		✓	0.903	0.875	0.759	1.737	0.699	0.650
		✓		✓		✓	0.904	0.807	0.958	0.935	0.875	0.714
	✓	✓		✓		✓	0.961	0.856	0.975	0.890	0.617	0.673
		✓		✓		✓	0.939	0.890	0.852	0.877	0.677	0.594
	✓	✓		✓	✓		0.955	0.835	0.961	0.898	0.784	0.649
✓		✓		✓		✓	0.953	0.853	0.965	0.921	0.617	0.708
	✓	✓	✓			✓	0.956	0.860	0.974	0.876	0.620	0.804
	✓	✓		✓		✓	0.962	0.891	0.975	0.832	0.601	0.634

Note: In terms of module selection, View1 is the module for learning SMILES representation, where “Single” denotes the use of a single-scale CNN-SE block and “Multi^c” denotes the use of the multiscale CNN-SE component; View2 is the module for learning fingerprint representation; View3 is the module for learning molecular graph representation, where “GCN” denotes the use of a graph convolutional neural network and “Multi^g” denotes the use of the multiscale GNN encoder; Fusion method is the module for fusing the feature information of different views, where “Concat” denotes the use of a concatenation operation, and “ATT” denotes the use of the dual cross-attention component. The symbol “✓” indicates that the module is used in the model. The best performance is marked in bold.

performance of MvMRL using convolution with kernel sizes of 2, 4, 8, and the combination of these three convolutional kernels (multiscale CNN-SE component), respectively. As shown in Fig. 5, the combination of these three convolutional kernels outperforms those of single-scale convolutions with kernel sizes of 2, 4, or 8 on all datasets. This indicates that the combination of the three convolutional kernels does improve the model’s performance. This may be due to the fact that the combination of three convolutional

kernels can integrate local dependencies, thus capturing features of SMILES sequences well.

Exploring the fusion order of different views

In this section, we analyze the fusion order of different molecular views, including the views from SMILES sequences, molecular fingerprints, and molecular graphs, which can offer valuable

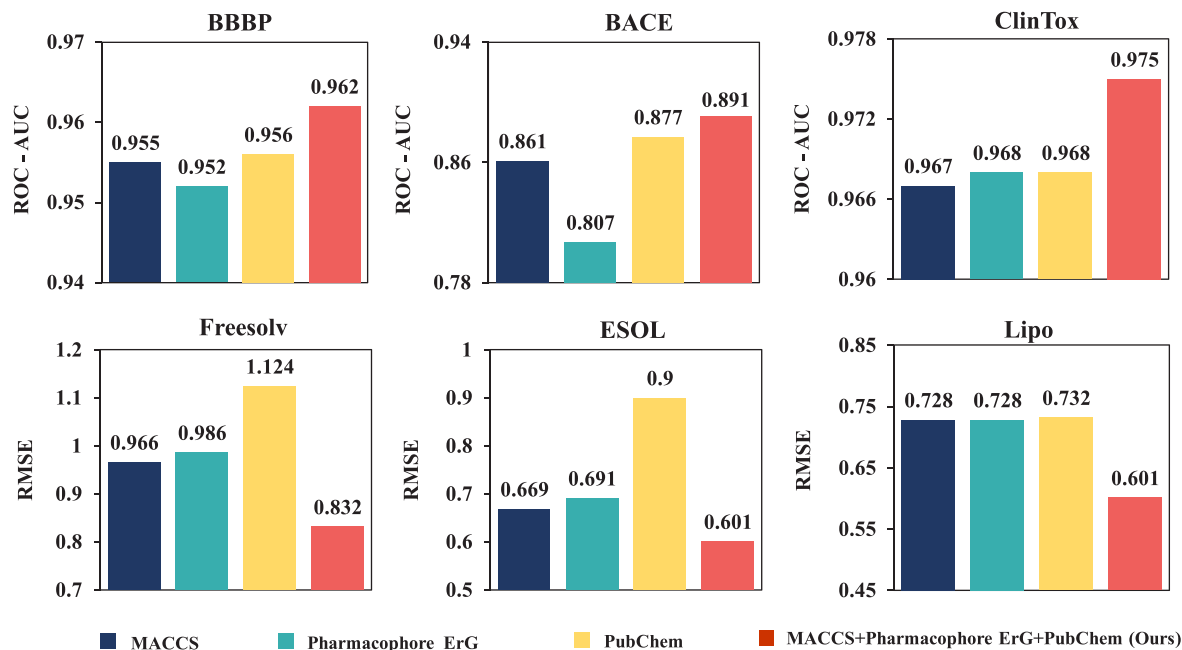


Figure 4. Results of different types of molecular fingerprints.

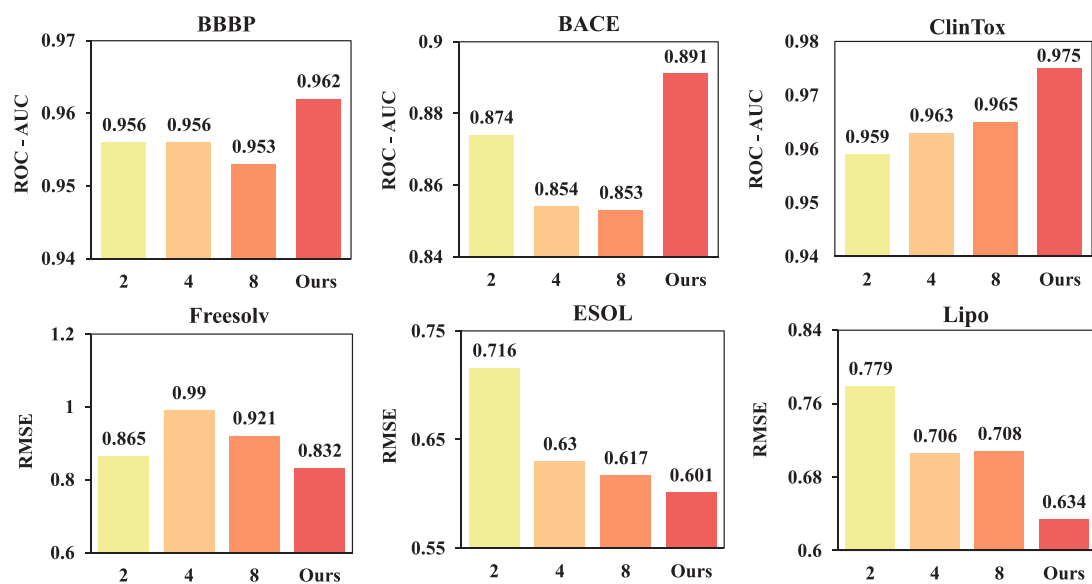


Figure 5. Results of different sizes of the convolutional kernel.

insights into exploiting the advantages of each perspective effectively. Since different views provide unique and complementary information, the order of their fusion is crucial for accurate and robust molecular property prediction.

In MvMRL, we design a specific fusion order, combining the View2 (for fingerprints) and the View1 (for SMILES sequences), the View2 and the View3 (for molecular graphs), respectively, and then concatenating the results of the previous two parts. In this section, we construct two variant model of MvMRL using two different fusion orders of different molecular views, including $MvMRL_{Concat(Att(v2,v1), Att(v3,v1))}$ and $MvMRL_{Concat(Att(v1,v3), Att(v2,v3))}$ to investigate their impacts on the model performance. From Table 7, we observed that the fusion order in which different views are integrated significantly impacts the model performance. Specifically, the MvMRL with the fusion order ($MvMRL_{Concat(Att(v1,v2), Att(v3,v2))}$) achieves the best performance,

implying that combining View1 and View2, View3 and View2 by cross attention, respectively, and then concatenating the results of the previous two parts can provide a more comprehensive representation of molecules that is better suited for molecular property prediction. This may be because molecular fingerprints provide additional molecular information not learned by the molecular graphs and SMILES sequences alone.

Exploring the effects of different aggregation methods

In this section, we use four aggregation methods, including Add, Mean, Max, and Concatenation (Concat), to explore the effects of different aggregation methods for the cross-attention operation of View2 with View1 and View3. Figure 6 and Table 8 show the results of different module aggregation methods on classification tasks and regression tasks. It can be observed that, overall, the

Table 7. Results of different fusion order of different views

Model	BBBP	BACE	ClinTox	Tox21	Sider	FreeSolv	ESOL	Lipo
MvMRL _{Concat(Att(v2,v1), Att(v3,v1))}	0.950	0.847	0.957	0.824	0.637	0.872	0.637	0.661
MvMRL _{Concat(Att(v1,v3),Att(v2,v3))}	0.956	0.881	0.953	0.795	0.581	0.900	1.276	0.680
MvMRL	0.962	0.891	0.975	0.842	0.653	0.832	0.601	0.634

Note: MvMRL_{Concat(Att(vi,vj), Att(vk,vl))} refers to the variant model of MvMRL that the model first combines the view i (v_i) and the view j (v_j) by cross attention (Att $(\cdot)$), the view k (v_k) and the view l (v_l), respectively, and then concatenating (Concat $(\cdot)$) the results of the previous two parts (Att (v_i, v_j) , Att (v_k, v_l)).

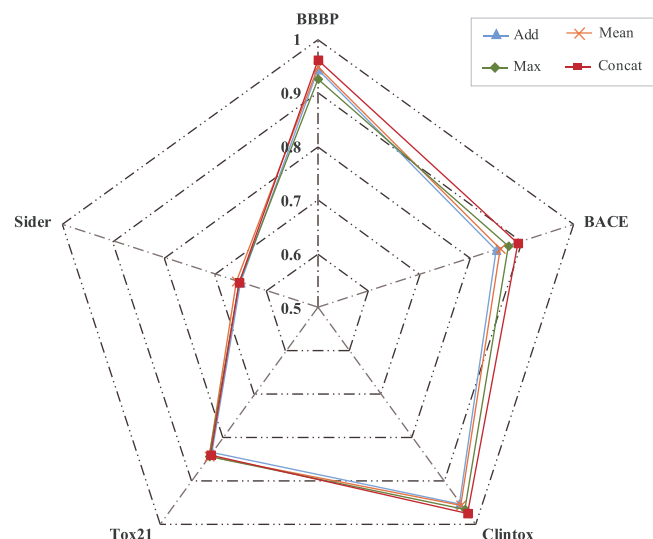


Figure 6. Results of different module aggregation methods on classification tasks.

Table 8. Results of different module aggregation methods on regression tasks

Aggregation method	FreeSolv	ESOL	Lipo
Add	1.568	0.675	0.751
Mean	1.209	0.681	0.688
Max	1.890	0.709	0.737
Concat (Ours)	0.832	0.601	0.634

result based on the Concat method is better than other aggregation methods. This suggests that Concat can better integrate information from the result of cross-attention of “View2” with “View1” and “View3.” Other aggregation methods are occasionally better than Concat due to the bias of certain datasets, resulting in variation in performance. On balance, Concat is still the optimal choice.

Exploring the visualization results of molecular representation

To better explore our model and the interpretability of the predicted results, following Zhu et al. [30], we visualize the learned representation of molecules on BBBP, BACE, ESOL, and Lipo datasets. Figure 7 shows the visualization results of molecular representation. On BBBP and BACE datasets, molecules are roughly divided into two clusters. On ESOL and Lipo datasets, there exists clear gradation from upper and lower left to middle right. It also confirms that our MvMRL could extract meaningful latent representations concerning the property and is interpretable.

Discussion

This section provides additional observations and discussion. The good performance of MvMRL could be explained by several factors. First, the multi-view learning framework of MvMRL contributes to its good performance. By considering multiple views of the molecules, MvMRL can capture different aspects and perspectives of molecular information, resulting in more comprehensive representations of the molecules. Second, MvMRL can capture both local and global information from molecular graphs and SMILES, as well as hybrid fingerprint information. By integrating three types of feature information, MvMRL can capture fine-grained detail information. Furthermore, the architecture of MvMRL contains cross-attention mechanisms, which allows the model to learn complex nonlinear relationships in the data. This flexibility enables MvMRL to adapt to different tasks and datasets.

In summary, the combination of the multi-view learning framework, the cross-attention mechanism, and the ability to capture both local and global features results in the good performance of MvMRL.

Conclusion

In this work, we proposed a novel method for multi-view molecular representation learning called MvMRL exploiting molecular information from SMILES, graphs, and fingerprints for molecular property prediction. We further considered the relationship between molecular information from different views. We used two multiscale feature extractors to obtain local information and global information from molecular graphs and SMILES, while using an MLP network to learn complex nonlinear relationships of molecular fingerprints. In addition, we developed a dual cross-attention component to interact with the features from three views. Extensive experiments on six benchmark datasets demonstrate the effectiveness of our MvMRL, which outperforms most state-of-the-art results.

Overall, our model can extract a more comprehensive molecular representation, demonstrating the great potential of multi-view molecular representation. However, the interpretability of the model is weak and our model does not perform the best on certain datasets such as the Sider and Lipo datasets. In the future, we will study how to improve the interpretability of the model and how to fuse 3D geometric information for more comprehensive learning of molecular information.

Key Points

- MvMRL uses the complementary information of molecular fingerprint, SMILES, and molecular graph representations, resulting in improving performance.

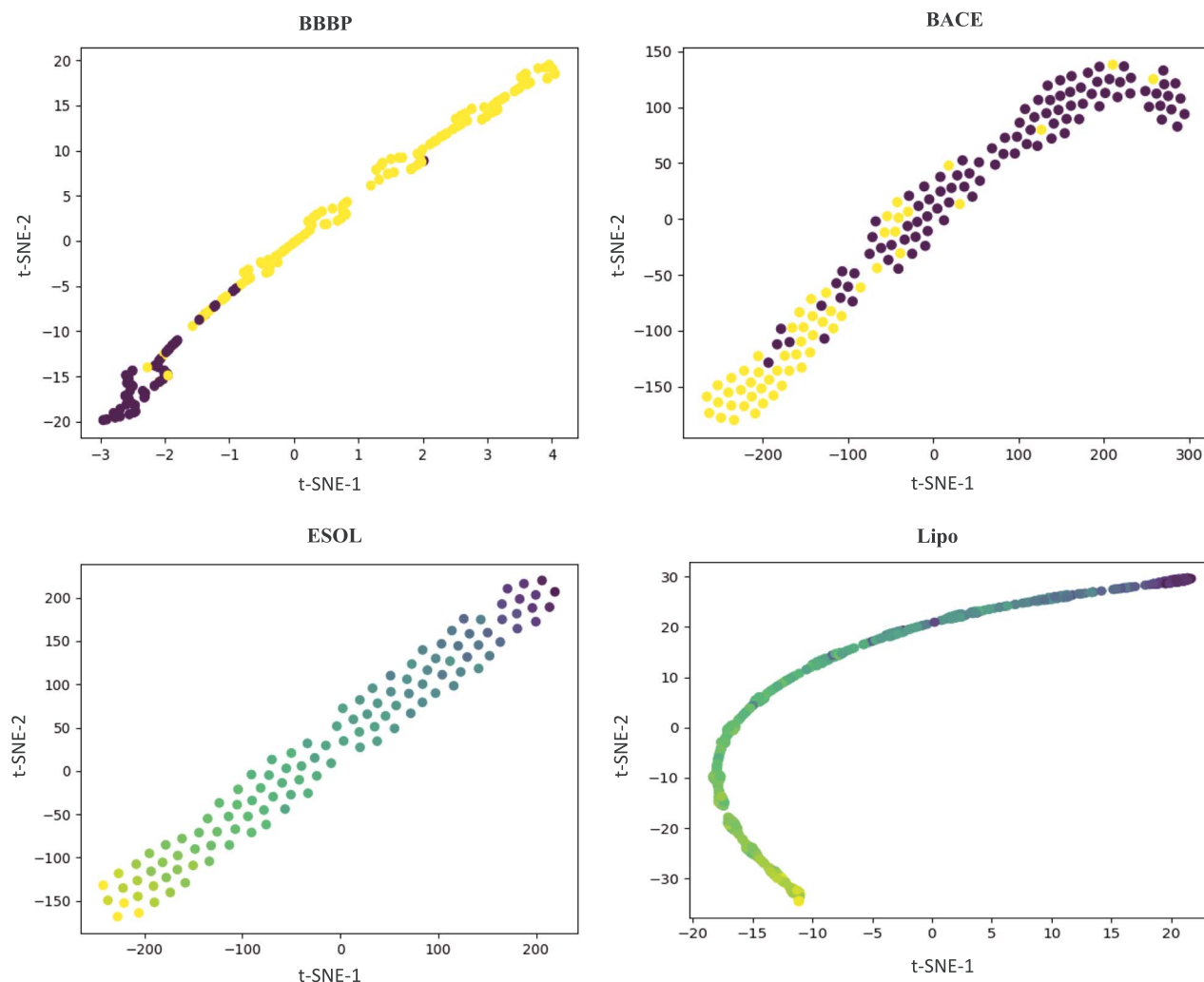


Figure 7. Molecular representation visualization on BBBP, BACE, ESOL, and Lipo dataset.

- MvMRL exploits two multiscale feature extractors for handling SMILES and graph representations, allowing for local-global interaction between them.
- MvMRL exploits a dual cross-attention component, which enables the model to concentrate on the crucial features between multiscale and nonlinear relationships.
- MvMRL outperforms state-of-the-art methods in a series of molecular property prediction, and ablation study demonstrates that our proposed model is effective.

Author contributions

Ru Zhang conceived and designed the study. Ru Zhang and Yijia Wu conducted the experiments. Ru Zhang performed the analyses and wrote the manuscript. Yuzhong Peng and Yanmei Lin supervised the study, performed the analyses, and revised the manuscript. Hao Zhang, Mingzhi Liao, and Lei Deng revised the manuscript. Yanmei Lin, Hao Zhang, Mingzhi Liao, and Yuzhong Peng are the co-corresponding authors.

Supplementary data

Supplementary data are available at *Briefings in Bioinformatics* online.

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Data availability

Data and codes in our experiments are released in <https://github.com/jedison-github/MvMRL>.

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